

Methodology

Overall Research Strategy

This project uses an empirical evaluation approach. The main goal is to test whether an LLM-based agent can identify missing data mechanisms when given datasets with known missingness patterns.

The study consists of three main stages:

1. Dataset selection

The study has three main stages: Dataset selection We select a set of classical datasets from the UCI Machine Learning Repository, which contains hundreds of widely used datasets from various fields, like healthcare, finance, and social science.

2. Missing data construction

For each dataset, we either:

- use datasets with existing missing values (if available), or
- introduce controlled missingness (MCAR, MAR, MNAR)

Each mechanism is applied several times with different missing rates and random seeds, creating about 100 datasets per mechanism.

3. Agent evaluation

The agent analyzes each dataset and identifies the missing data mechanism, providing an explanation.

Data Sources and Limitations

The main data source is the UCI Machine Learning Repository, a popular resource for comparing machine learning methods.

We focus on “classic” datasets because:

- they are well-documented
- they are frequently used in research
- they allow reproducibility

However, there are limitations:

- many datasets are relatively clean
- true missingness mechanisms are often not documented
- some preprocessing is required

To tackle this, we combine real datasets with synthetically generated missingness, providing both realism and control.

Key Assumptions

This study makes several assumptions:

1. Simulated missingness is a reasonable proxy

We assume that standard methods for generating MCAR, MAR, and MNAR reflect real-world patterns.

2. **Model outputs reflect reasoning**

We assume that the agent's explanations give insight into its reasoning, although this may not always be the case.

3. **Classic datasets are sufficient for evaluation**

While they are widely used, they might not capture all real-world complexities.

These assumptions are common but will be acknowledged as limitations.

Experimental Design

We construct datasets as follows:

- 10–20 base datasets
- Missing rates: 10%–50%
- Multiple random seeds
- Three mechanisms: MCAR, MAR, MNAR

This results in:

- ~100 datasets per mechanism
- ~300 total experimental cases

For each dataset, the agent is asked to:

- describe the missing data pattern
- identify the mechanism
- explain its reasoning

We record both predictions and explanations.

Evaluation Approach

We assess the agent using three types of metrics:

1. **Classification accuracy**

This checks if the agent correctly identifies MCAR, MAR, or MNAR.

2. **Consistency across prompts**

We test the same dataset under different prompt conditions (e.g., neutral vs. misleading) to see if answers change.

3. **Reasoning quality**

We evaluate qualitatively whether the explanation shows a correct statistical understanding. This combined approach helps us tell apart:

- correct reasoning
- and correct answers by chance or prompt bias

Feasibility and Risk Management

This project faces several challenges:

Data availability

Data availability Not all datasets have known missingness mechanisms.

→ Solution: combine real and synthetic missingness

MNAR difficulty

MNAR is inherently hard to verify

→ Solution: use standard proxy methods and acknowledge limitations

Evaluation subjectivity

Evaluating reasoning isn't fully objective.

→ Solution: define simple and consistent criteria

Time constraints

Large-scale experiments may take time.

→ Solution: automate dataset generation and evaluation

Likely Objection and Response

A likely concern is:

Synthetic missing data may not reflect real-world behavior.

This is a valid point. To address it, we adopt a hybrid approach:

- real datasets for realism
- synthetic missingness for control

This balance lets us focus on the agent's behavior while keeping practical feasibility.